Cloud Cost Model – Two-Tier Web Service

1. Introduction:

This document estimates the monthly cloud infrastructure cost for a two-tier web service under three operating scenarios: Lean, Balanced, and Resilient.

The model considers application compute, database, storage, network transfer, logging and monitoring, backups, and support. The budget limits and expected users are based on the provided SLO and cost-envelope worksheet.

Note: The cost allocations below are illustrative planning estimates, not live cloud-provider quotations. Actual costs depend on the cloud provider, region, service configuration, usage, and applicable taxes.

2. Scenario Overview:

Parameter	Lean	Balanced	Resilient
Monthly Budget	₹2,500	₹8,000	₹25,000
Expected Users	500	5,000	25,000
Availability Target	99%	99.5%	99.9%
p95 Latency Target	800 ms	500 ms	300 ms
RTO	4 hours	1 hour	30 minutes
RPO	24 hours	1 hour	15 minutes

3. Monthly Cost Breakdown:

The following allocations show one possible way to distribute each scenario's budget across the main infrastructure components.

Cost Component	Lean	Balanced	Resilient
Application Compute	₹700	₹2,000	₹6,000
Database	₹600	₹1,800	₹5,000
Storage	₹200	₹600	₹1,500
Network Transfer	₹250	₹800	₹3,000
Logging and Monitoring	₹200	₹700	₹2,000
Backup and Recovery	₹250	₹900	₹4,000
Support and Contingency	₹300	₹1,200	₹3,500
Total Monthly Budget	**₹2,500**	**₹8,000**	**₹25,000**

These figures are budget allocations for planning purposes. They should be validated against actual provider pricing before deployment.

4. Annual Budget Estimate:

Annual estimates are calculated by multiplying the monthly budget by 12.

Scenario	Monthly Budget	Estimated Annual Budget
Lean	₹2,500	₹30,000

Balanced	₹8,000	₹96,000
Resilient	₹25,000	₹3,00,000

These estimates assume the monthly budget remains constant throughout the year. They exclude unexpected usage increases, price changes, and taxes unless included in the selected budget.

5. Scenario Design:

Lean Scenario

- Designed for approximately 500 expected users.
- Uses a small application deployment and cost-effective database configuration.
- Uses basic monitoring and scheduled backups.
- Targets 99% availability.
- Suitable for prototypes, demonstrations, and low-traffic applications.

Balanced Scenario

- Designed for approximately 5,000 expected users.
- Provides more application and database capacity.
- Includes regular monitoring, backups, and recovery procedures.
- Targets 99.5% availability.
- Suitable for applications with moderate traffic and growing usage.

Resilient Scenario

- Designed for approximately 25,000 expected users.
- Allocates more budget to compute, database capacity, monitoring, and recovery.
- Requires stronger backup and recovery arrangements to support the tighter RTO and RPO targets.
- Targets 99.9% availability.
- Suitable for higher traffic and services that require improved reliability.

The exact infrastructure design must be tested to confirm that it can meet the stated availability, latency, and recovery targets.

6. Cost Optimization Strategies:

1. **Right-size compute:** Select instance sizes based on measured CPU and memory usage.
2. **Optimize the database:** Use appropriate indexes, connection pooling, and query optimization.

3. **Control storage:** Remove unnecessary files and apply suitable retention policies.
4. **Reduce network costs:** Compress responses and avoid unnecessary data transfers.
5. **Manage logs:** Retain operational logs for an appropriate period and avoid excessive logging.
6. **Review backups:** Balance backup frequency, retention, and recovery requirements.
7. **Monitor spending:** Set budget alerts and review usage regularly.
8. **Scale when needed:** Use autoscaling where supported and appropriate for the workload.

Cost reductions should not compromise security, data integrity, or required recovery objectives.

7. Assumptions and Limitations:

- The three monthly budget limits come from the provided planning worksheet.
- The component-level cost allocations are illustrative estimates and are not verified market prices.
- Actual costs vary by provider, region, compute capacity, database configuration, storage volume, traffic, and support plan.
- Free-tier credits and discounts are not assumed.
- Taxes and one-time migration costs are not separately included.
- The expected user counts are planning inputs, not guaranteed capacity limits.
- Load testing, cost monitoring, and provider price checks are required before production deployment.

8. Conclusion:

The Lean, Balanced, and Resilient scenarios provide three budget options for different user loads and reliability requirements. The monthly budgets are ₹2,500, ₹8,000, and ₹25,000 respectively. Actual infrastructure costs and performance should be validated before selecting a production configuration.